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THE UNIVERSITY OF HONG KONG

COMP4801

Final Year Project

Interim Report

Aggregating Epigenetic Methylation Signals for Improved Long-Read Genetic Variant Calling

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Abstract

Contemporary haplotype-aware variant callers predominantly rely on heterozygous Single Nucleotide Polymorphisms (SNPs) for read phasing. However, the scarcity of SNPs in homozygous or repetitive genomic regions compromises phasing continuity, resulting in fragmented haplotypes and suboptimal variant calling accuracy. To address this limitation, this project introduces a novel Clair3-Methylation workflow that leverages Allele-Specific Methylation (ASM) signals as an orthogonal data source. The proposed algorithm identifies high-confidence ASM sites using Fisher's Exact Test to assign haplotypes to previously unphased reads and concatenates adjacent phase sets based on consistent methylation signatures. Preliminary benchmarking on Genome in a Bottle (GIAB) datasets (10x-50x coverage) demonstrates substantial improvements in haplotype contiguity. Notably, the Phase Set N50 on chromosome 20 increased from ~0.9 Mb to ~33 Mb at 50x coverage. These results confirm that integrating epigenetic markers effectively bridges genomic gaps left by SNP-only methods, leading to superior recall and F1-scores in complex genomic contexts.

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Abbreviations

Abbreviation	Full Name
5hmC	5-hydroxymethylcytosine
5mC	5-methylcytosine
ASM	Allele-Specific Methylation
BAM	Binary Alignment Map
InDel	Insertions/Deletions
LRS	Long-read Sequencing
NGS	Next Generation Sequencing
ONT	Oxford Nanopore Technologies
PacBio	Pacific Biosciences
SNP	Single Nucleotide Polymorphism
SV	Structural Variation
VCF	Variant Call File

1. Introduction

1.1 Background

Genetic variants are the differences in DNA sequence relative to a reference genome [1]. These variants encompass a wide spectrum, from single nucleotide polymorphisms (SNPs) and small insertions/deletions (InDels) to large-scale structural variations (SVs) [1]. Their accurate analysis is critical for understanding human genetic diversity, diagnosing Mendelian disorders, developing targeted cancer therapies, and assessing disease risk [2]. The proliferation of Long-read Sequencing (LRS), led by Oxford Nanopore Technologies (ONT) and Pacific Biosciences (PacBio) [3], has greatly improved the detection of these variants at a genome-wide scale, as it offers a powerful solution to many of the limitations imposed by short-read technologies, also known as Next Generation Sequencing (NGS) [4]. By generating reads that can be tens to hundreds of thousands of basepairs in length, these technologies can span entire repetitive elements and provide the long-range contextual information required to resolve complex genomic architectures and unambiguously map reads [3].

This capability has spurred the development of a new class of sophisticated variant calling tools to find variants, known as "haplotype-aware" variant callers. Fundamentally, these tools are designed based on the biological principle of diploidy, meaning humans carry two copies of each chromosome—one maternal and one paternal. The specific combination of variants inherited together on a single chromosome copy is defined as a 'haplotype'. Unlike traditional methods that assess each genomic position in isolation, haplotype-aware callers operate on a more biologically faithful principle. By evaluating how well each read supports pre-constructed haplotypes, the software can effectively distinguish true genetic variants (which appear consistently on one haplotype) from random sequencing errors and systematically call SNPs and InDels simultaneously [5]. Prominent examples utilizing this approach include pipelines such as Clair3 or PEPPER-Margin-DeepVariant [5][6].

1.2 Project Motivation

However, virtually all contemporary haplotype-aware callers primarily rely on a single type of genetic marker: the heterozygous SNP [5][6]. A heterozygous SNP is defined as a genomic position where two parental chromosomes have different alleles. The process of phasing, the critical first step of partitioning sequencing reads into two haplotypes, relies on identifying reads that cover these known heterozygous sites. However, the phasing

algorithm is deprived of its necessary input in genomic regions where heterozygous SNPs are sparse or entirely absent [7]. These challenging areas include the homozygous regions, low complexity regions, and repetitive regions containing tandem repeats or segmental duplications. This leads to a cascade of negative consequences: 1. phase blocks become fragmented and short, 2. reads that do not span a heterozygous site cannot be assigned a haplotype [7]. Consequently, variant calling accuracy and completeness plummet precisely in these genomic contexts that are already the most challenging to analyze [5].

To overcome these issues, this project utilizes an orthogonal data type that is not subject to the same constraint: DNA methylation. DNA methylation is an epigenetic modification, a chemical tag added to DNA that can alter gene expression without changing the underlying sequence [8]. With Long-read Sequencing Technologies like ONT, this information can be simultaneously detected in a basepair resolution [8], without the need for destructive chemical treatments [9], thus can serve as a new data dimension in genetic variant calling. In particular, the utilization of this epigenetic information is facilitated by a unique pattern known as Allele-Specific Methylation (ASM). ASM refers to the phenomenon where methylation patterns differ between the two chromosomes (i.e., alleles) inherited from the paternal and maternal sides [10], which correspond to the two haplotypes that must be partitioned. This enables ASM as a epigenetic marker to serve as a robust haplotype marker even in the absence of nearby heterozygous SNPs [7]. This approach represents a novel multi-omic analysis paradigm that integrates epigenomics with genomics, a direction justified by recent methods such as NanoMethPhase, which use phased haplotypes to phase genome-wide 5mc from LRS, even at low sequencing coverages [11].

1.3 Project Objectives

This project seeks to enhance the accuracy, completeness, and contiguity of haplotype-aware variant calling, especially in difficult regions like the homozygous regions, low complexity regions, and high repetitiveness regions. The objective of this project is to design, implement, and rigorously validate a novel computational method that integrates ASM signals in the genetic variant calling.

1.4 Project Scope and Deliverables

This project focuses on the variant calling performance improvement over a baseline SNP-only phasing method across a suite of key metrics, including precision, recall, and F1-score

for both SNPs and overall variants.

This project will deliver a Clair3-methylation variant calling workflow, which is comprised of the following parts:

- An algorithm that identifies ASM sites from ONT LRS data that contains simultaneous sequence and methylation information. This information is stored in Binary Alignment Map (BAM) files, the standard format for storing aligned sequencing data, as MM and ML tags [12]. Two methylation types, including 5-methylcytosine (5mC) and 5-hydroxymethylcytosine (5hmC), are taken into account [10], [13].
- A method that leverages these identified ASM "epigenetic markers" to assign definitive haplotype identity to sequencing reads that were previously unphased due to a lack of spanning SNP information. This will increase the percentage of haplotagged reads. We will also create a module that concatenates adjacent, but previously disconnected phase sets based on consistent ASM patterns observed in overlapping reads, thereby increasing the length and contiguity of reconstructed haplotypes. Better phasing performance can be observed in BAM files.

1.5 Report Outline

The structure of this report is organized as follows: Section 2 is the methodology and details the procedures and justifications for the technologies and algorithms we propose. Section 3 discusses current progress and presents preliminary experimental results. Section 4 discusses some difficulties encountered and outlines future directions. Finally, Section 5 concludes the report by summarizing the whole project.

2. Methodology

The methodology section is structured into four sections. In section 2.1, the framework of our pipeline is introduced. In section 2.2, the data preparation procedure is discussed. In section 2.3, a detailed algorithm design is presented. Finally in section 2.4, a set of comprehensive evaluation methods is proposed.

2.1 Pipeline Framework

This project is implemented as a bioinformatics pipeline primarily written in Python,

leveraging established and well-documented scientific libraries such as pysam for efficient manipulation of sequencing alignment files, scipy for statistical computations, and pandas for data management. It is integrated with the advanced Clair3 variant calling model to become a Clair3-Methylation workflow (see Figure 1).

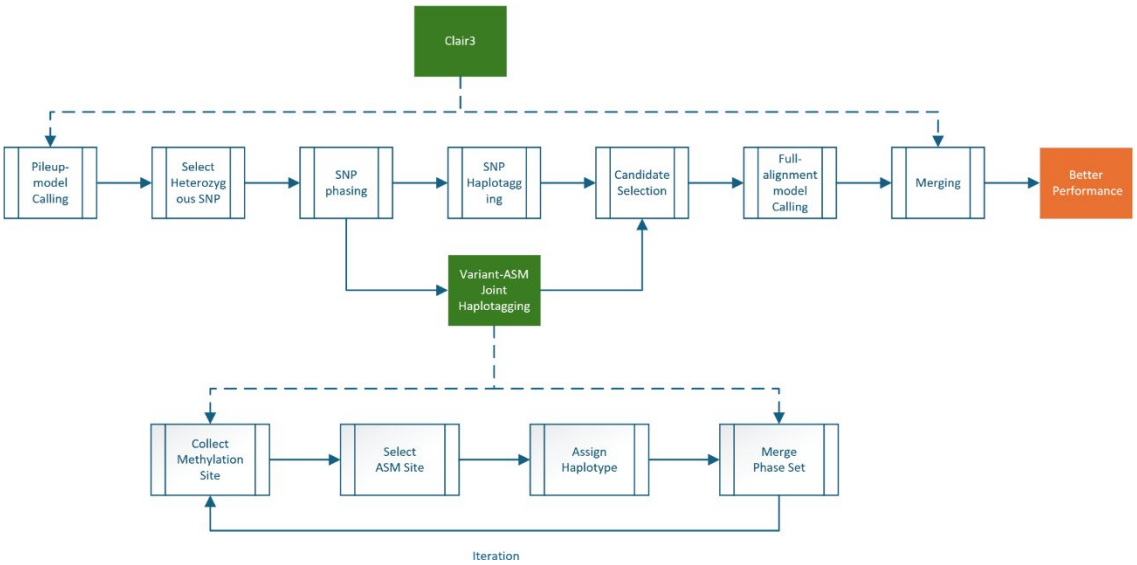

Figure 1: Proposed Clair3-Methylation workflow. The standard Clair3 pipeline (top) is augmented by our Variant-ASM Joint Haplotagging module (bottom), which iteratively identifies ASM sites to improve read phasing and phase set merging before final variant calling.

2.2 Data Acquisition and Preparation

The primary dataset for the development, testing, and evaluation is the publicly available Genome in a Bottle (GIAB) whole-genome sequencing dataset [14]. This dataset is the industry standard for benchmarking variant calling methods, as it is accompanied by a high-confidence "truth" variant call set against which performance can be accurately measured [14]. For sequencing data, ONT LRS data is selected, which provide more affordable and even longer reads than PacBio [15]. To assess the robustness of the method, datasets at various sequencing coverages (e.g., 10x, 20x, 30x, 40x, 50x) should be analyzed. The initial data preparation step involves aligning the raw reads to the GRCh38 human reference genome using a long-read aligner to produce the primary input for the pipeline: a coordinate-sorted BAM file containing methylation tags.

2.3 Algorithm Implementation

As illustrated in Figure 1, the workflow augments the standard Clair3 pipeline (top flow) with the proposed 'Variant-ASM Joint Haplotagging' module (bottom flow). While Clair3

performs its own SNP-based phasing, the new module independently collects methylation data to select ASM sites [10]. These sites are then used to assign haplotypes to previously unphased reads and to merge disconnected phase sets. This enhanced phasing information is then utilized by the Clair3 full-alignment model to improve the accuracy of the final variant calls.

2.3.1 Interpreting the Input

This subsection corresponds to the ' Collect Methylation Site ' step shown in Figure 1. The algorithm's input is a BAM file generated by a long-read sequencer, where the reads have already been haplotagged by a standard tool (e.g., WhatsHap). This file contains reads assigned to Haplotype 1 (HP:i:1), Haplotype 2 (HP:i:2), and a subset of unphased reads (lacking an HP tag). Methylation information are stored in MM/ML tags in the BAM file. MM shows the location of the 5mC and 5hmC, and ML is a 0-255 number showing the confidence that a 5mC or 5hmC exists. ML values are binarized into 0/1 methylation calls using a threshold selected by a criterion such as Gini impurity.

2.3.2 ASM Site Identification

This subsection corresponds to the ' Select ASM Site ' step shown in Figure 1. The algorithm identifies ASM sites through each existing phase set (a contiguous genomic region with a group of phased reads). Within each set, it systematically scans for CpG dinucleotides that exhibit a bimodal methylation pattern, where a substantial fraction of covering reads are methylated and a substantial fraction are unmethylated [10]. The reason we only focus on CpG dinucleotides is that they are the locations where ASM prevails [9]. For each candidate site, a 2x2 contingency table (Table A. 1 in the appendix) is constructed. Fisher's Exact Test is then applied to this table. The test determines if there is a statistically significant non-random association between haplotype assignment and methylation status (e.g., p-value < 0.05). A significant result validates the site as a high-confidence ASM locus, where methylation status acts as a reliable proxy for haplotype identity. This statistical test is chosen for its accuracy in analyzing categorical data, especially when cell counts may be small, as is common in sequencing read data.

2.3.3 Epigenetic Haplotype Assignment

This subsection corresponds to the ' Assign Haplotype ' step shown in Figure 1. After identifying a catalog of high-confidence ASM sites, the algorithm performs a second pass over all previously unphased reads that cover these loci. If an unphased read displays a methylation pattern that is strongly characteristic of one of the defined haplotypes (e.g., the read is methylated at a site where Haplotype 1 is predominantly methylated and Haplotype

2 is not), it is confidently assigned to that haplotype by adding the appropriate HP tag. This step directly recovers phasing information that was lost by SNP-only methods.

2.3.4 Phase Set Concatenation

This subsection corresponds to the ' Merge Phase Set ' step shown in Figure 1. The algorithm inspects the boundaries of adjacent but disconnected phase sets. If two neighboring sets are found to share a consistent ASM signature on reads that span the gap between them (e.g., reads assigned to Haplotype 1 are methylated in both sets at a linking ASM site), the algorithm merges them into a single, longer, and more contiguous phase block. This directly addresses the critical goal of increasing overall haplotype length and contiguity.

2.4 Evaluation and Benchmarking

The output of our algorithm is a phased BAM file. The final output of the whole integrated pipeline is a variant call file (VCF), the standardized text file format for storing gene sequence variations. The accuracy of this VCF is rigorously benchmarked against the GIAB high-confidence truth set using standard evaluation tools like hap.py [16]. The evaluation is designed to be multifaceted to ensure biologically and clinically significance, focusing on two distinct but related areas of improvement:

1. **Phasing Improvement:** The primary metrics include the absolute and percentage increase in the number of reads assigned an HP tag in the BAM file, demonstrating the direct recovery of phasing information. Furthermore, another measurement is the haplotype contiguity using the phase block N50 statistic, a standard measure where 50% of the entire assembly is contained in phase blocks of this size or larger.
2. **Variant Calling Accuracy:** Standard performance metrics—precision, recall, and F1-score—will be calculated for both SNPs and InDels. A key component of the analysis is to stratify these results by genomic context to demonstrate that the most significant improvements occur in regions known to be challenging for standard methods.

3. Progress and Results

This section presents the project progress and results. In section 3.1, a brief discussion of progress so far is given. In section 3.2, currently achieved preliminary results are presented, providing an insight of the final full-scale benchmarking.

3.1 Project Progress

This project is divided into three phases. Phase 1 conducts literature review on the background information and existing tools. Phase 2 develops the core algorithm, proposing a first version in selecting ASM sites and utilizing them. It tries to seek an increase in both phasing performance and final variant calling performance. Phase 3 finalizes the algorithm and endeavours to achieve significant gain in SNP calling recall and F1 score, especially in difficult regions. Phase 1 and 2 have been finished in October and November. The current algorithm can process TBs of data end-to-end without errors in acceptable time, and successfully demonstrates its performance to a certain extent. Figure A. 1 in the appendix is an example of two selected ASM coexisting with an SNP. Phase 3 has not started yet due to time limitation and will be left to Semester 2, 2025-2026.

3.2 Preliminary Results

This part discusses results currently achieved. To assess the performance of this project, two sets of metrics are proposed. Section 3.2.1 and 3.2.2 are the analysis for phasing performance, measuring the direct results of our algorithm. Section 3.2.3 and 3.2.4 are the analysis for variant calling performance, justifying this project's contribution on downstream tasks.

3.2.1 Haplotagged Read Percentage

Haplotagged read percentage is the most direct metric that justify the effectiveness of this project. It measures the percentage of haplotagged reads compared to total reads. As shown in Figure A. 2 in the appendix, the proposed method consistently outperforms the baseline SNP-only method across all chromosomes. From 10x to 50x, haplotagged read percentage shows a uniform increase of 5-10%, with 50x CHR16 has a 12.52% increase. This uniform improvement indicates that the inclusion of ASM sites successfully recovers phasing information for reads that do not span heterozygous SNPs. It suggests that methylation signals are sufficiently abundant and distributed to complement SNP sparsity, thereby rescuing reads that would otherwise remain unphased.

3.2.2 Phase Set Number and Phase Set N50

To assess haplotype contiguity, we analyzed the total number of phase sets and the Phase Set N50 metric, as presented in Figure A. 3 and Figure A. 4 in the appendix, respectively. Figure A. 3 demonstrates a widespread reduction in the number of phase sets across all chromosomes when using our pipeline. Concurrently, Figure A. 4 reveals a substantial

increase in the Phase Set N50 length. Most notably, at 50x coverage, the Phase Set N50 for chromosome 20 surges from approximately 935 kb in the baseline to over 33 Mb in our workflow. The reduction in phase set count combined with the exponential growth in N50 confirms the effectiveness of the 'Phase Set Concatenation' module. This indicates that our algorithm is correctly identifying ASM signatures to merge adjacent blocks, creating a far more continuous and reliable haplotype assembly along the genome. The correlation between higher coverage and larger N50 gains further suggests that deep sequencing provides the necessary density of methylation signals to bridge larger genomic gaps.

3.2.3 Overall Variant Calling Performance

The impact of improved phasing on the final variant calling accuracy is summarized in Table A. 1. The table compares Precision, Recall, and F1-scores between the standard Clair3 pipeline and our methylation-enhanced workflow. At 30x coverage, the proposed method increases the total F1-score from 0.979627 to 0.980672. Crucially, the number of False Negatives (FN) is reduced significantly, dropping from 118,193 to 108,171. These metrics demonstrate that the upstream improvements in phasing translate directly into downstream accuracy. The reduction in False Negatives is particularly significant, as it implies that the model is now detecting variants that were previously missed due to ambiguous phasing, thereby improving the overall sensitivity (Recall) of the caller without sacrificing precision.

3.2.4 Performance in Difficult Regions

To validate the method's performance in challenging genomic contexts, we stratified the F1-score improvements by region type, as visualized in Figure A. 5. The analysis reveals positive F1-score gains in “Low Complexity” regions, specifically within “TR201-10kbp” and “Homopol_gt11bp”. Conversely, the chart also highlights slight performance regressions (negative improvement) in “CDS” and “large Segmental Duplications (>10kb)”. The gains in low-complexity regions validate our core hypothesis: ASM signals serve as an effective substitute for SNPs in repetitive regions where alignment is difficult and SNPs are often absent. However, the performance drop in CDS suggests that our current ASM selection logic may struggle in regions with complex gene regulation features, identifying a specific area for algorithmic refinement in Phase 3 of the project.

4. Discussions

This section first identifies some difficulties encountered during the algorithm development

procedures and then provides a brief discussion on the future work to be accomplished.

4.1 Difficulties Encountered

During the algorithm development procedure, the following difficulties were encountered.

- **Ambiguity in Methylation Probability (ML) Definitions.** ML tags in BAM files come from Dorado Basecaller [11]. It is a 0-255 numerical value, but no information on how to interpret this value is provided. Section 2.3.1 thus provides a basic analysis on its distribution and translates it to 0/1 representation.
- **Signal Noise in Long-Read Methylation Data.** The quality of the methylation data is highly important to our algorithm as the detection of ASM sites is based directly on methylation signals. Currently, the noise in methylation data cannot be neglected [7]. To mitigate this, we implemented strict filtering criteria as described in Section 2.3.2.

4.2 Future Work

The immediate next steps will focus on finalizing the algorithm and completing the full-scale benchmark evaluation described in Section 2.4.

- **Complete Integration:** A more rigorous ASM site selection method can be proposed. Currently, in regions without surrounding SNP existence, the algorithms select ASM sites based on merely methylation ratio difference. This may not entirely exclude the influence of random methylation pattern.
- **Full-Scale Benchmarking:** Execute the complete Clair3-Methylation pipeline on the entire GIAB whole-genome dataset and repeat for all available coverages (10x, 20x, 30x, 40x, 50x) to test robustness. Use hap.py [16] to compare the VCF output against the GIAB truth set. This will generate the final precision, recall, and F1-score metrics for both SNPs and InDels.

While it’s beyond the scope of this project, the long term work could focus on the development of a novel "Epigenome-Aware Variant Caller". This could be achieved by integrating all types of methylation information directly as a feature channel into the deep learning model of current variant callers.

5. Conclusion

This project is motivated by the inherent limitations of contemporary haplotype-aware

variant callers, which rely heavily on heterozygous SNPs for read phasing. In genomic regions where these SNPs are sparse—such as homozygous or low-complexity regions—standard algorithms fail to maintain haplotype connectivity, resulting in reduced variant calling accuracy. The primary objective of this project is to design and validate a computational workflow that utilizes Allele-Specific Methylation as an orthogonal phasing signal. To achieve this, we developed a Clair3-methylation pipeline that integrates a novel module for ASM site identification using Fisher's Exact Test, followed by an haplotype assignment and phase set concatenation algorithm.

The preliminary benchmarking on the GIAB dataset has demonstrated that epigenetic markers can effectively substitute for missing genetic markers. The most notable finding is the dramatic improvement in haplotype contiguity; specifically, the Phase Set N50 on chromosome 20 increased from ~0.9 Mb to over 33 Mb at 50x coverage. This indicates that our phase set concatenation module successfully bridged gaps between previously fragmented blocks. Consequently, this improved phasing translated into a tangible enhancement in variant calling performance, achieving higher F1-scores and lower false negative rates compared to the SNP-only baseline.

The significance of this work lies in its proof-of-concept for "multi-omic" variant calling. By successfully leveraging methylation data natively available in ONT sequencing, this project provides a robust solution for resolving complex genomic structures that are traditionally "dark" to sequence-only analysis. It establishes that biological signals beyond the DNA sequence can be computationally harvested to improve the resolution of the human genome.

Despite these successes, the study identified specific limitations. The algorithm currently relies on a basic thresholding approach for methylation, which may be susceptible to signal noise inherent in current upstream tasks. Furthermore, a slight performance regression is observed in large segmental duplications, suggesting that the current ASM selection criteria may yield false positives in regions with complex methylation heterogeneity.

Future work must address these limitations to ensure clinical-grade accuracy. It is recommended that subsequent work incorporate a more rigorous filtering mechanism to distinguish true ASM from random noise. Additionally, the long-term evolution of this project should move towards a fully "Epigenome-Aware" model, where methylation data is ingested as a direct feature channel rather than a post-processing heuristic. Immediate next steps will involve a full-scale benchmark across all GIAB datasets to confirm robustness.

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Appendices

	Methylated Reads	Unmethylated Reads
Haplotype 1	Count A	Count B
Haplotype 2	Count C	Count D

Table A. 1: Contingency table for Fisher's Exact Test

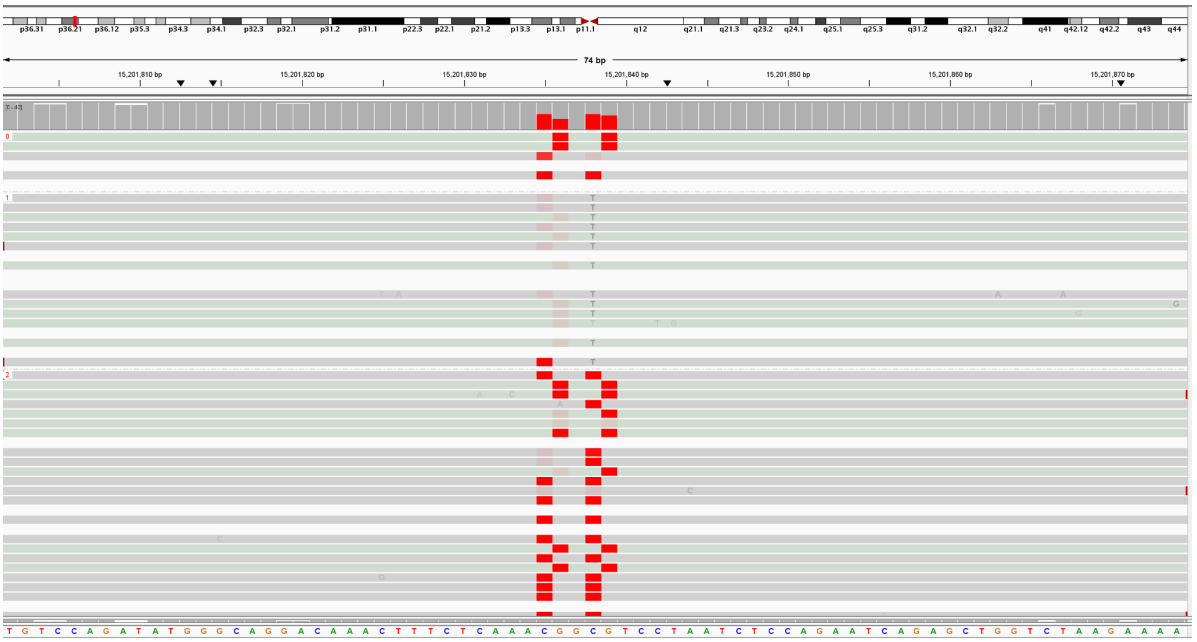

Figure A. 1: Screenshot of two ASM sites around a variant. Here haplotype 2 is methylated and haplotype 1 is unmethylated. The algorithm successfully detected a genetic variation at location 15201838.

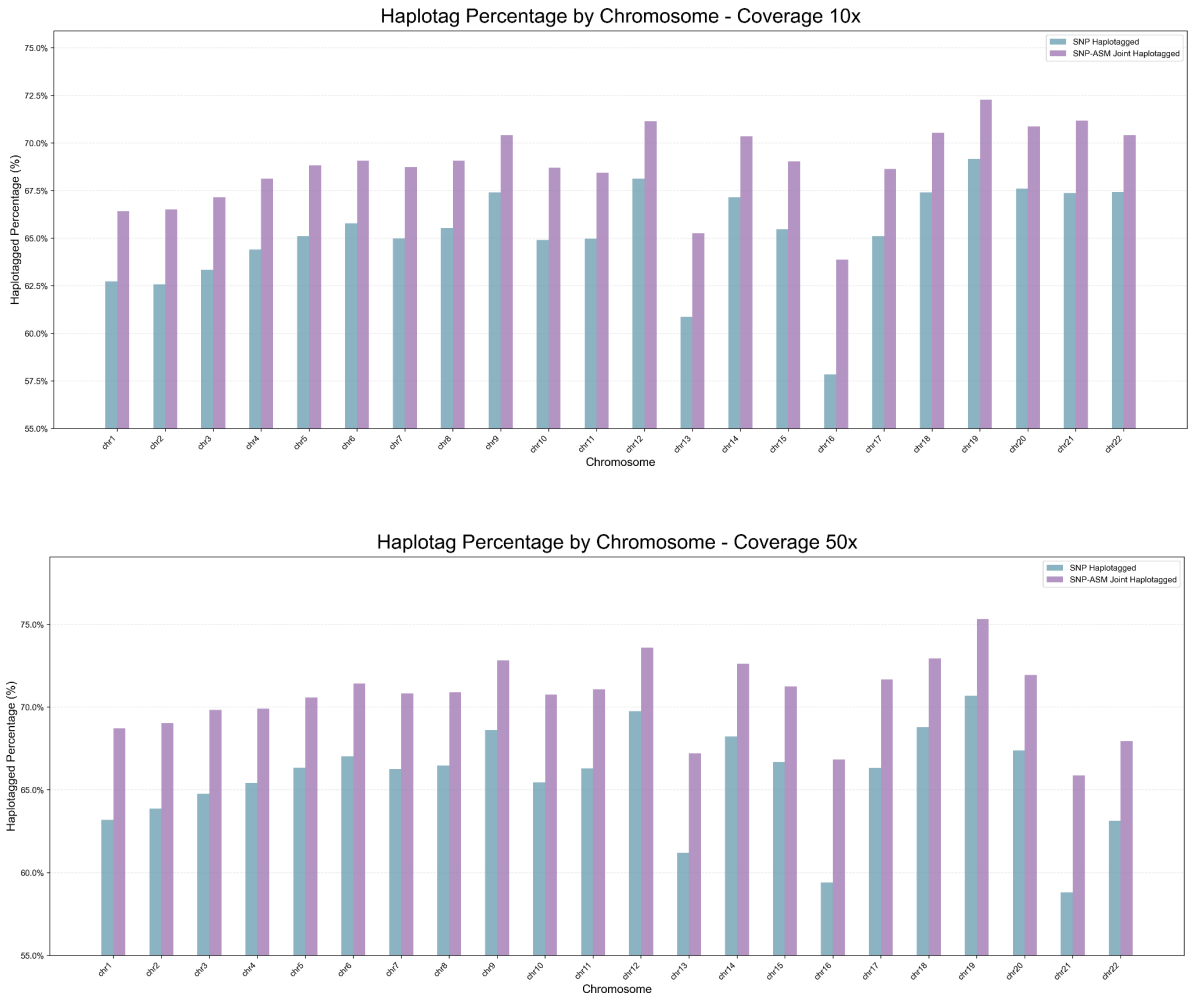

Figure A. 2: Haplotagged Read Percentage Increase by Chromosome (10x and 50x coverage). From 10x to 50x, haplotagged read percentage show a uniform increase of around 6-10% across chromosomes, with 50x CHR16 has a 12.52% increase.

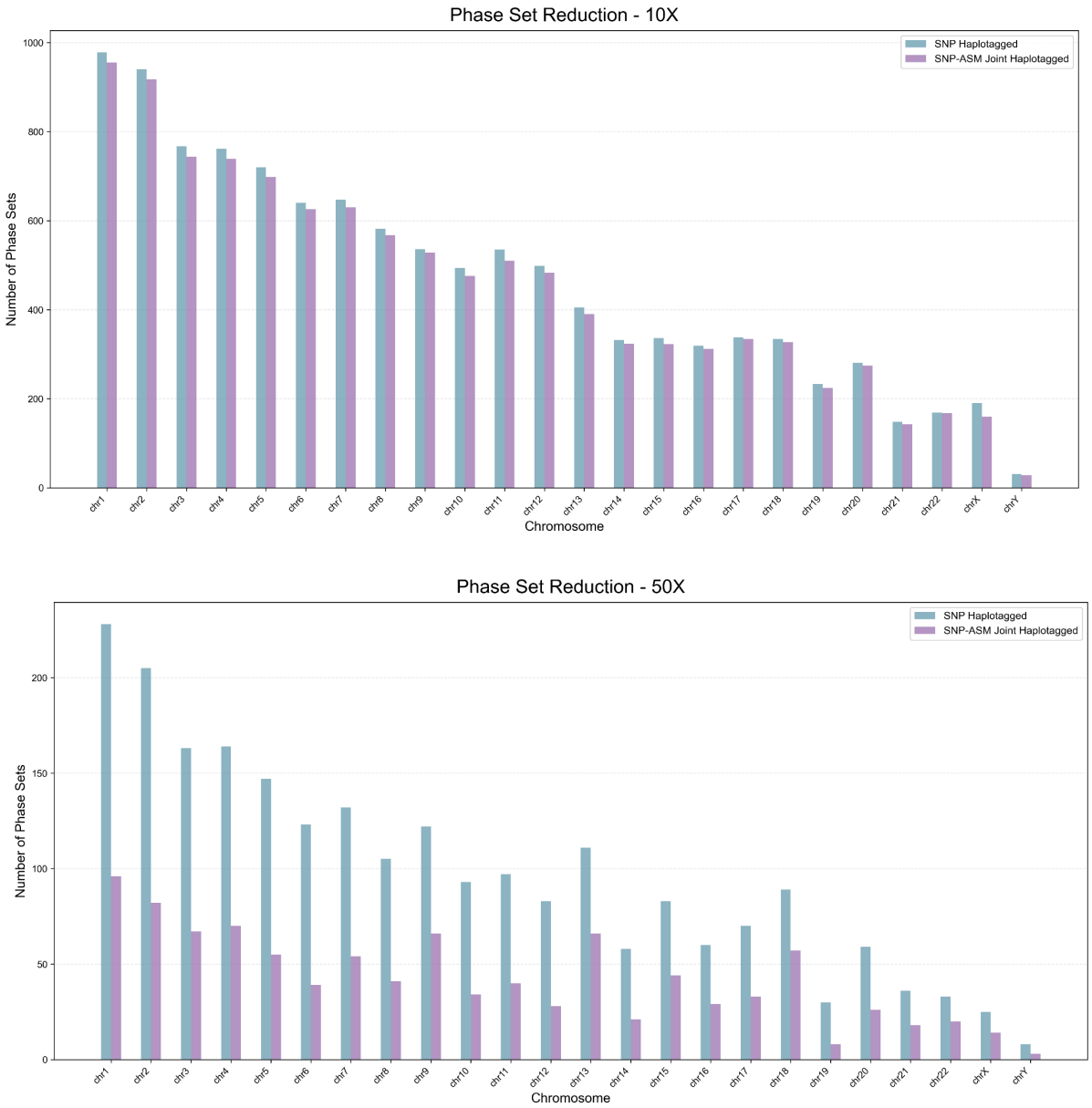

Figure A. 3: Phase Set Count Decrease by Chromosome (10x and 50x coverage). The decrease is more significant in larger coverages.

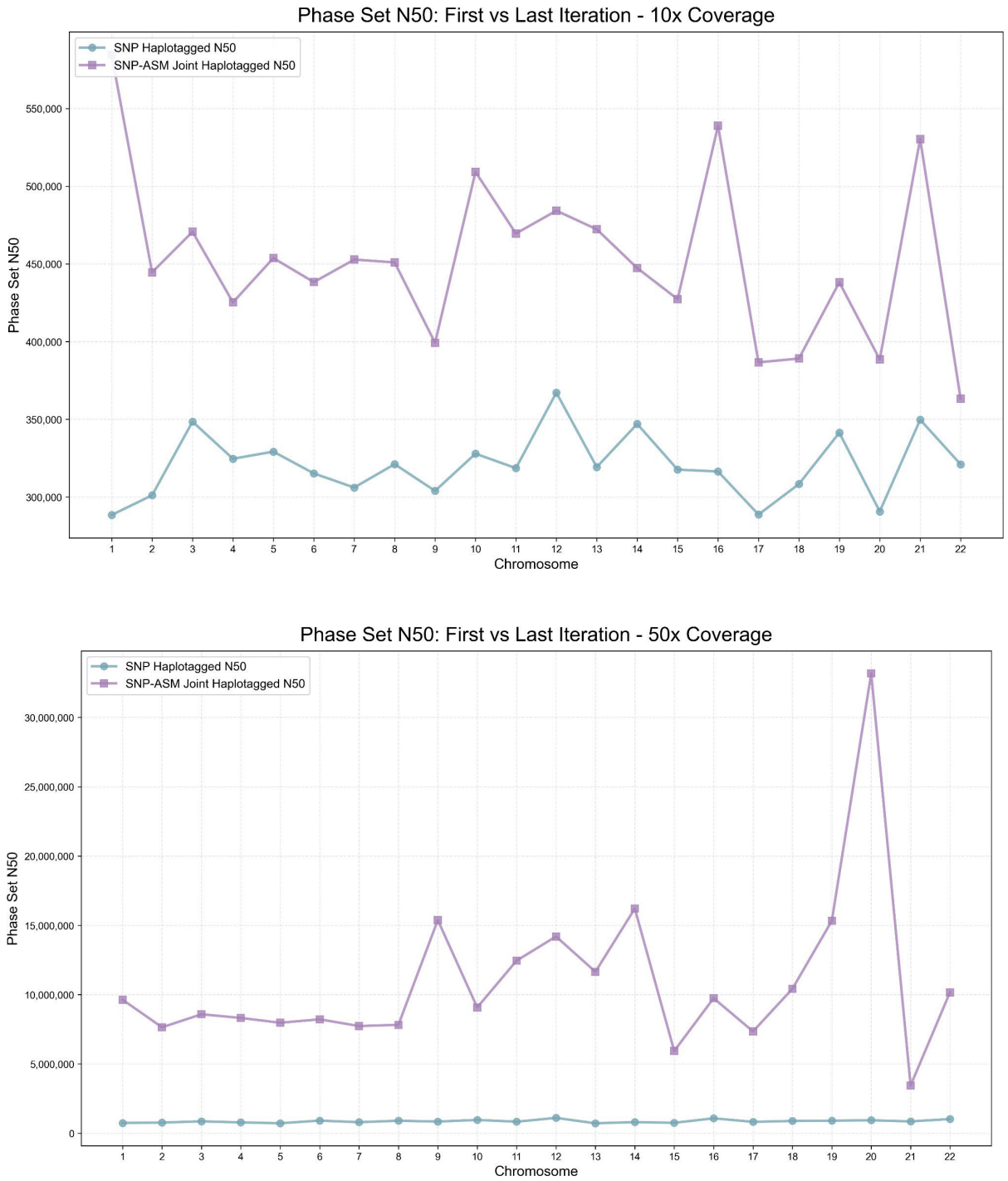

Figure A. 4: Phase Set N50 Increase by Chromosome (10x and 50x coverage). Remarkable increase is uniform across chromosomes.

Platform	Dataset	Caller	SNP						Total					
			Precision	Recall	F1-score	TP	FP	FN	Precision	Recall	F1-score	TP	FP	FN
ONT 5kHz	HG003 20x	Clair3	0.994576	0.991975	0.993274	3,300,793	18,010	26,702	0.985264	0.962068	0.973528	3,686,641	55,319	145,355
		Clair3 with implemented methylation	0.994131	0.99405	0.99409	3,307,695	19,535	19,800	0.984064	0.964438	0.974152	3,695,723	60,043	136,273
ONT 5kHz	HG003 30x	Clair3	0.997397	0.994162	0.995777	3,308,068	8,638	19,427	0.990325	0.969156	0.979627	3,713,803	36,412	118,193
		Clair3 with implemented methylation	0.997419	0.996669	0.997044	3,316,411	8,584	11,084	0.989737	0.971772	0.980672	3,723,825	38,754	108,171
ONT 5kHz	HG003 40x	Clair3	0.998218	0.994715	0.996464	3,309,910	5,911	17,585	0.992041	0.97272	0.982286	3,727,459	30,019	104,537
		Clair3 with implemented methylation	0.998102	0.997213	0.997657	3,318,221	6,314	17,585	0.991536	0.975356	0.983379	3,737,561	32,029	94,435
ONT 5kHz	HG003 50x	Clair3	0.998509	0.995477	0.996991	3,312,444	4,947	15,051	0.992768	0.975674	0.984147	3,738,780	27,346	93,216
		Clair3 with implemented methylation	0.998563	0.997469	0.998016	3,319,072	4,778	8,423	0.992496	0.977809	0.985098	3,746,960	28,444	85,036

Table A. 2: Clair3 variant calling results compared with methylation refined results, which use the result VCF of Clair3 as input of ASM-SNP Joint Haplotagging, then run Clair3 full alignment model

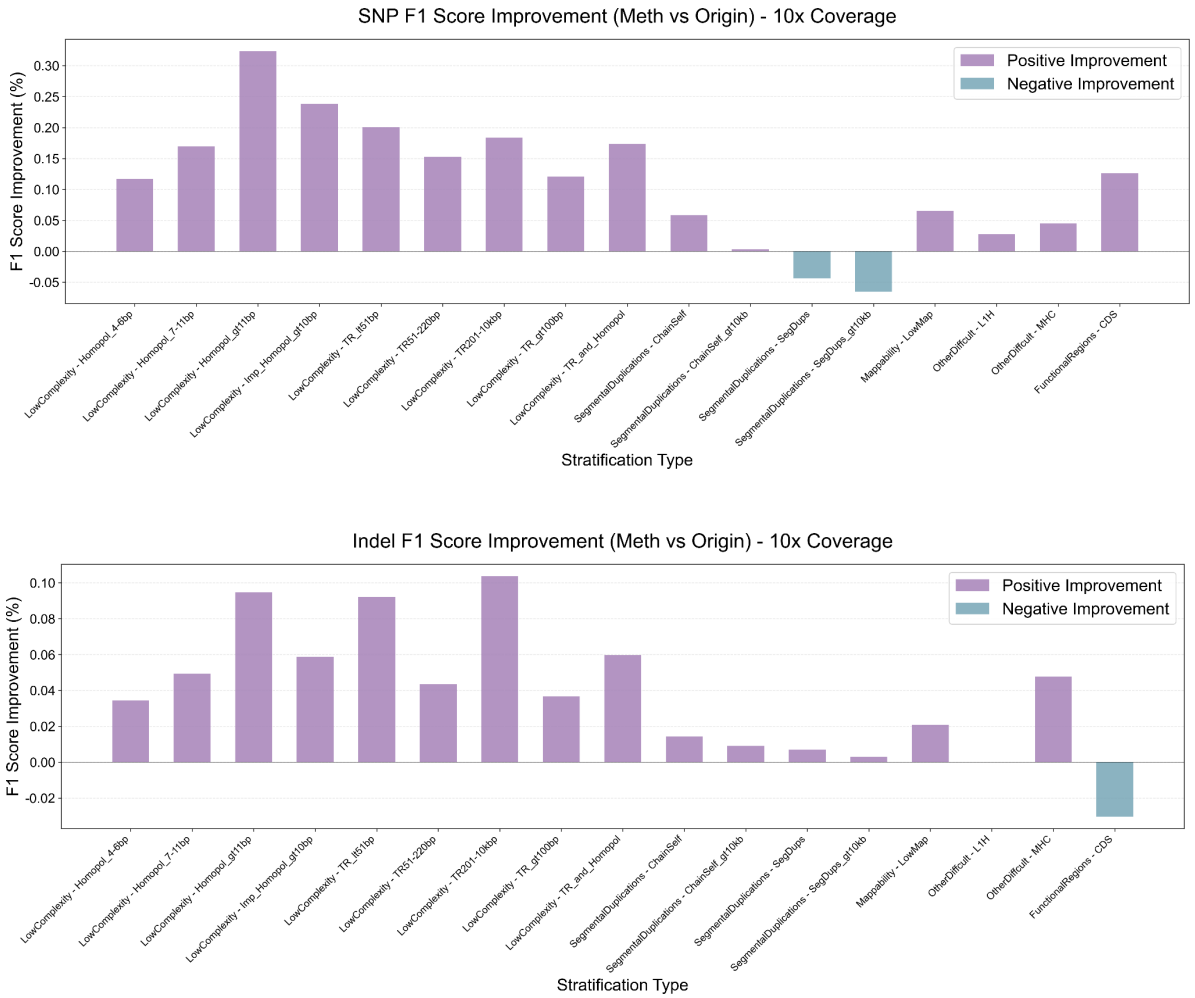

Figure A. 5: The F1 score improvement for SNP and InDel (10x) in Difficult Regions. Positive improvements are observed in most difficult regions.